

## Quiz 5

Filename: No File Submission

### Instructions

All answers must be submitted on Webcourses. Before proceeding with the quiz on Webcourses, you are expected to first work through and answer the questions here and on paper. Only once you have completed your work should you proceed to Webcourses and submit your responses.

Please note that some questions may not require you to submit your final written answers; however, you are still expected to fully solve every question on your own. As you work through the problems, you must time yourself. Each question is designed to be completed within a reasonable amount of time. Some questions may take approximately 1 minute, others around 5 minutes, but no question should take more than 10 minutes.

### TODO 1 (10 points)

The code segment below computes a value stored in the variable `sum`. Express the final value of `sum` as a nested summation. Do not simplify or solve the summation.

```
int sum = 0;

for(int i = 1; i < 100; i++)
    for(int j = 1; j <= i; j++)
        sum += (2*i+j*j);

printf("sum = %d.\n", sum);
```

### TODO 2 (20 points)

Determine a closed-form solution for the summation shown below, in terms of  $n$ . Express your answer in standard polynomial form  $an^2 + bn + c$ , where  $a$ ,  $b$  and  $c$  are constants.

$$\sum_{i=1}^n \left( \sum_{j=1}^{3i+7} 2 \right)$$

### TODO 3 (20 points)

Consider applying Merge Sort to the array shown below. What does the array look like immediately before the final call to the Merge function is executed?

**Original Array**

|       |    |    |   |    |    |    |   |    |   |    |
|-------|----|----|---|----|----|----|---|----|---|----|
| index | 0  | 1  | 2 | 3  | 4  | 5  | 6 | 7  | 8 | 9  |
| value | 37 | 15 | 9 | 17 | 22 | 64 | 2 | 25 | 6 | 13 |

**Array right before last call to Merge**

|       |   |   |   |   |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|---|---|---|---|
| index | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| value |   |   |   |   |   |   |   |   |   |   |

#### TODO 4 (10 points)

The array shown below represents the result after exactly one call to the `Partition` function (i.e., the first partition step in Quick Sort). Based on the partitioned array, identify the pivot element (i.e., the value of the data). Then, justify your answer by giving three valid reasons that confirm why your chosen element must be the pivot.

|       |    |    |    |    |    |    |    |    |    |
|-------|----|----|----|----|----|----|----|----|----|
| index | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| data  | 17 | 20 | 14 | 13 | 10 | 27 | 50 | 34 | 48 |

### TODO 5 (10 points)

**Note:** If you answered this question before April 1, the version on Webcourses has been updated. Please go back and answer the revised question again, as the answer choices have been updated.

Consider the following recursive function. Write the recurrence relation for the given function in terms of  $T(n)$ . You do not need to solve the recurrence relation.

```
int compute(int n) {  
    if(n == 1)  
        return 2;  
    else  
        return 2 * compute(n-1) + 3;  
}
```

### TODO 6 (30 points)

Use the iteration technique to determine the Big-Oh run-time of the following recurrence relation.

$$T(n) = 2T\left(\frac{n}{4}\right) + 5 \qquad T(1) = 1$$

#### Hint

You may find the following formula useful while solving this:  $a^{\log_a n} = \sqrt[r]{n}$